Beloiu Tudor-Andrei

Bucharest, Romania

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Education

University of Bucharest — **Oct. 2024 – Jul. 2027**
B.Sc. in Computer Science — *Bucharest, Romania*

- GPA Year 1: **9.00/10**
- **Merit Scholarship** recipient for outstanding academic performance in Year 1

"Ecaterina Teodoroiu" National College — **Sep. 2020 – Jul. 2024**
Baccalaureate in Mathematics & Computer Science — *Targu Jiu, Romania*

- Baccalaureate Grades: **Mathematics 10/10, Computer Science 10/10**

Projects

Industrial Anomaly Detection | *Python, PyTorch, ResNet18* — **github.com/tudorbeloiu/anomaly-detection**

- Built a **Convolutional Autoencoder (CAE)** trained on normal-only images, using a combined **MSE + SSIM loss** to preserve structural patterns in reconstructions.
- Analyzed per-category failure modes across all 15 MVTec AD categories and implemented a **pretrained ResNet18 feature extractor** with **Mahalanobis distance** scoring for categories where the autoencoder failed.
- Achieved **0.95+ AUROC** on texture categories (grid, wood, hazelnut) and **0.88+ AUROC** on complex objects (cable, metal_nut) by selecting the best method per category.

Traffic Light Color Classifier | *Python, PyTorch, OpenCV, Pandas* — **github.com/tudorbeloiu/TrafficLights**

- Built a custom **Convolutional Neural Network (CNN)** to classify traffic light colors (red, yellow, green), designed as an accessibility tool for colorblind drivers.
- Handled class imbalance for the yellow class using weighted **Cross Entropy Loss**, reaching 95% test accuracy.
- Used **YOLOv8** to detect and crop traffic lights from real street images, feeding them into the CNN for color classification.

TeamFinder | *Java, Spring Boot, React, PostgreSQL, AWS S3* — **www.teamfinder.ro**

- Built and deployed a full-stack collaboration platform where developers post projects, apply to join teams, and communicate privately after acceptance.
- Implemented **JWT** authentication stored in HttpOnly cookies with token blacklisting on logout and bcrypt password hashing.
- Secured auth endpoints with per-IP **rate limiting** using a token bucket algorithm (Bucket4j) and validated file uploads against magic bytes to prevent content-type spoofing.

User-Level Thread Library | *C, x86 Assembly, Linux API* — **github.com/tudorbeloiu/User-Level_Thread_Library**

- Developed a **preemptive scheduler** from scratch, implementing **Round-Robin** logic and custom context switching in x86 Assembly.
- Developed synchronization primitives (Mutex with Handoff, Semaphores) and a Thread Pool with a blocking Producer-Consumer queue.
- Implemented memory safety mechanisms using Stack Guard Pages to intercept stack overflows.

Achievements

Nitro NLP Hackathon | *7th place out of 73 teams* — **2026**

County Mathematics Olympiad | *Honorable Mention* — **2024**

County Informatics Olympiad | *Honorable Mention* — **2022**

Technical Skills

Languages: Python, Java, C++, C#, SQL, C, Assembly (x86)
Frameworks & Libraries: PyTorch, NumPy, Pandas, scikit-learn, Spring Boot, OpenCV, .NET, ASP.NET Core (MVC, REST API), React, Entity Framework Core, SFML
Tools & Dev: Git, Docker, Linux, Burp Suite, Wireshark, Ghidra, GDB, Makefiles
Spoken Languages: Romanian (Native), English (B2)